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Yikai Wu

Education

Princeton University — **Princeton, NJ, USA**
Ph.D. student in Computer Science — *2022–present*
Advisor: Prof. Sanjeev Arora

Yale University — **New Haven, CT, USA**
Ph.D. student in Computer Science — *2021–2022*

Duke University — **Durham, NC, USA**
Bachelor of Science (Summa Cum Laude) — *2017–2021*
Double major in Computer Science and Mathematics, Minor in Physics
Overall GPA: 3.97/4.00
Member of honor society Phi Beta Kappa

Research Experience

Difficult synthetic math question generation — **Princeton University**
Advisor: Prof. Sanjeev Arora — *Feb 2026–present*

- Ongoing project developing cost-efficient pipelines for constructing difficult synthetic math questions using multiple model calls for generation, solving, checking, refinement, and skill annotation. Generated questions with controlled skill combinations and filtered them using answer consistency across teacher models, with the goal of improving reasoning-data quality and evaluating model robustness.

Gradient-based skill analysis for model interpretability — **Princeton University**
Advisor: Prof. Sanjeev Arora — *Nov 2025–present*

- Ongoing project analyzing whether gradient updates of language models can reflect the mathematical skills used in solving math questions. Developed a gradient factor analysis pipeline that extracts layer-wise factors from high-dimensional LoRA gradient updates, then compresses them into interpretable skill representations using dictionary-learning ideas. We also annotate mathematical skills using frontier models and compare these annotations with the gradient-based analysis, with the broader goal of discovering model-internal skill structure without relying only on pre-defined skill lists.

Skill-based reinforcement learning for mathematical reasoning — **Princeton University**
Advisor: Prof. Sanjeev Arora — *Oct 2025–Feb 2026*

- Studied skill-targeted reinforcement learning and supervised fine-tuning for improving language models on mathematical reasoning. Used stronger language models to identify required skills and diagnose missing skills in incorrect solutions, then constructed training data targeting those weaknesses. Found that skill-targeted SFT was useful in related settings, while skill-targeted RL did not consistently outperform standard RL, suggesting that RL may already be relatively robust to some training-test distribution shifts.

Language-model agents for numerical program optimization **Princeton University**
Mentor: Dr. Ofir Press *Apr 2025–Oct 2025*

- Contributed to *AlgoTune*, an open-ended benchmark of 154 expert-curated numerical programming tasks with a framework for validating and timing language-model-generated code against reference implementations from packages such as SciPy, scikit-learn, and CVXPY.
- Evaluated *AlgoTuner*, a baseline LM agent that edits, compiles, profiles, verifies, and selects optimized code, achieving an average $1.72\times$ speedup over reference solvers while primarily finding implementation-level rather than algorithmic improvements.
- Paper [1] was presented at the Conference on Neural Information Processing Systems (NeurIPS) 2025.

Step-by-step Chain-of-Thought reasoning analysis **Princeton University**
Advisor: Prof. Sanjeev Arora *Feb 2025–Oct 2025*

- Studied whether Chain-of-Thought (CoT) prompting reflects genuine step-by-step reasoning or reliance on memorized patterns. We introduce step-by-step Chain-of-Thought (sbs-CoT), which decomposes reasoning into iterative steps across multiple forward passes, preventing reliance on memorized templates. We found universal performance degradation from CoT to sbs-CoT, non-linear scaling effects in reasoning robustness around 10–30B parameters, and three major failure modes: context loss, step fragmentation, and premature termination.

Benchmarking AI methods for combinatorial optimization **Princeton University**
Mentor: Prof. Sanjeev Arora *Oct 2023–Jan 2025*

- Benchmarked recent AI-inspired methods, including generative models and reinforcement learning approaches, against strong classical solvers for NP-hard combinatorial optimization problems, focusing on Maximum Independent Set.
- Showed that the classical solver KaMIS consistently outperforms leading AI-based methods, even on in-distribution random graphs, and introduced a serialization analysis explaining why some non-backtracking AI methods behave similarly to simple greedy heuristics.
- Paper [2] was published in Transactions on Machine Learning Research (TMLR), April 2026.

Differentially Private Optimization on Unitary Orbits **Yale University**
Mentor: Prof. Nisheeth Vishnoi *Jun 2021–Feb 2022*

- Consider the following optimization problem: Given $n \times n$ matrices A and Λ, maximize $\langle A, U\Lambda U\rangle$ where U varies over the unitary group $\mathrm{U}(\Lambda)$. The optimization can be used in multiple matrix approximation problems such as PCA and rank-k approximation. We designed an efficient differentially private algorithm for this optimization where A is constructed using users' private data. We also provided upper and lower bounds for the utility of the optimization.
- Paper [3] was presented at the Conference on Learning Theory (COLT) 2022.

Hessian in Neural Networks **Duke University**
Mentor: Prof. Rong Ge *May 2020–Aug 2021*

- The Hessian of the loss function captures important properties of neural networks. We observed that the Hessian has several interesting structures which appear commonly in different neural networks. We explained these structures using Kronecker factorization. Our new understanding of the Hessian can be used with PAC-Bayes techniques to get better generalization bounds.
- Paper [4] on arXiv.

Differential Privacy for Multiple Analysts **Duke University**
Mentor: Prof. Ashwin Machanavajjhala *Jan 2020–May 2021*

- Differential privacy is the gold standard of privacy protection. Existing differentially private algorithms are designed for a single analyst and are problematic for multiple analysts. We formulated three criteria to decide whether an algorithm is good for multiple analysts. We demonstrated how existing algorithms fail to satisfy them. We also designed new differentially private algorithms which satisfy these criteria provably.
- Paper [5] was presented at the International Conference on Very Large Data Bases (VLDB) 2021. The preliminary version was presented at the Theory and Practice of Differential Privacy Workshop (TPDP) 2020.

Differential Privacy for Summation Queries **Duke University**
Mentor: Prof. Ashwin Machanavajjhala *May 2018–Nov 2019*

- Answering summation queries under differential privacy is a little understood, non-trivial problem. Traditional differentially private algorithms for these queries are data-independent and often result in large errors for some types of data. We proposed a data-dependent algorithm using truncation to effectively reduce the errors in the results, while providing the same level of privacy protection.
- Paper [6] was presented at the Theory and Practice of Differential Privacy Workshop (TPDP) 2019.

Quantum Information and Algorithms **Duke University**
Mentor: Prof. Iman Marvian *May 2019–May 2020*

- Quantum information applies modern physics knowledge to develop a new type of computers and communication devices. We investigated a type of quantum operators useful for ion-trap quantum computers. We also designed and analyzed algorithms to purify quantum states, which is useful for quantum communication and quantum error correction.
- In addition, I studied several research papers on quantum algorithms, quantum learning theory, and quantum complexity theory.

Publications

[1] Ori Press, Brandon Amos, Haoyu Zhao, Yikai Wu, Samuel Ainsworth, Dominik Krupke, Patrick Kidger, Touqir Sajed, Bartolomeo Stellato, Jisun Park, et al. Algotune: Can language models speed up general-purpose numerical programs? *Advances in Neural Information Processing Systems*, 38, 2025.
Presented at the Conference on Neural Information Processing Systems (NeurIPS) 2025.

[2] Yikai Wu, Haoyu Zhao, and Sanjeev Arora. Unrealized expectations: Comparing AI methods vs classical algorithms for maximum independent set. *Transactions on Machine Learning Research*, 2026. ISSN 2835-8856. URL `https://openreview.net/forum?id=ksGoCT5zW6`.
Published at the Transactions on Machine Learning Research (TMLR) April 2026.

[3] Oren Mangoubi, Yikai Wu, Satyen Kale, Abhradeep Thakurta, and Nisheeth K Vishnoi. Private matrix approximation and geometry of unitary orbits. In *Conference on Learning Theory*, pages 3547–3588. PMLR, 2022.
Presented at the Conference on Learning Theory (COLT) 2022.

[4] Yikai Wu, Xingyu Zhu, Chenwei Wu, Annie Wang, and Rong Ge. Dissecting Hessian: Understanding common structure of Hessian in neural networks. *arXiv:2010.04261 [cs.LG]*, 2020.

[5] David Pujol, Yikai Wu, Brandon Fain, and Ashwin Machanavajjhala. Budget sharing for multi-analyst differential privacy. *Proceedings of the VLDB Endowment (PVLDB)*, 14(10): 1805–1817, 2021. doi: 10.14778/3467861.3467870.
Presented at the International Conference on Very Large Data Bases (VLDB) 2021.

[6] Yikai Wu, David Pujol, Ios Kotsogiannis, and Ashwin Machanavajjhala. Answering summation queries for numerical attributes under differential privacy. *arXiv:1908.10268 [cs.DB]*, 2019.
Presented at the Theory and Practice of Differential Privacy Workshop (TPDP) 2019.

Honors and Awards

Computing Research Association (CRA)
Outstanding Undergraduate Researcher Honorable Mention *2020*

Mathematical Contest in Modeling (MCM)
Meritorious Winner *2019*

Duke University Mathematics Student Award
The Karl Menger Award *2018*

The International Collegiate Programming Contest (ICPC)
Mid-Atlantic Regional Ranked 4th *2017, 2018*

William Lowell Putnam Mathematical Competition
Ranked 142.5 (Top 4 at Duke University) *2017*

Technical Skills

Python, PyTorch, verl, SFT, RLVR, LLM evaluation, synthetic data generation, C, C++, R, Mathematica